

Uber Supply Demand Gap Analysis

Project Objective

Analyze Uber request data to identify demand-supply gaps, Peak hours, cancellation Patterns and provide business recommendations

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings("ignore")
```

Load Dataset

```
In [2]: df = pd.read_csv("Uber Request Data.csv")
df.head()
```

```
Out[2]:
```

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
0	619	Airport	1.0	Trip Completed	11/07/16 11:51	11/07/16 13:00
1	867	Airport	1.0	Trip Completed	11/07/16 17:57	11/07/16 18:47
2	1807	City	1.0	Trip Completed	12/07/16 9:17	12/07/16 9:58
3	2532	Airport	1.0	Trip Completed	12/07/16 21:08	12/07/16 22:03
4	3112	City	1.0	Trip Completed	13/07/16 8:33	13/07/16 9:25

4 – Dataset Information

```
In [4]: df.info()
df.shape
df.describe(include='all')
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6745 entries, 0 to 6744
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Request id            6745 non-null   int64
1   Pickup point          6745 non-null   object
2   Driver id             4095 non-null   float64
3   Status                6745 non-null   object
4   Request timestamp     6745 non-null   object
5   Drop timestamp        2831 non-null   object
dtypes: float64(1), int64(1), object(4)
memory usage: 316.3+ KB

```

Out[4]:

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp
count	6745.000000	6745	4095.000000	6745	6745	2831
unique	NaN	2	NaN	3	4016	2282
top	NaN	City	NaN	Trip Completed	15/07/16 19:19	13/07/16 8:53
freq	NaN	3507	NaN	2831	8	5
mean	3384.644922	NaN	149.501343	NaN	NaN	NaN
std	1955.099667	NaN	86.051994	NaN	NaN	NaN
min	1.000000	NaN	1.000000	NaN	NaN	NaN
25%	1691.000000	NaN	75.000000	NaN	NaN	NaN
50%	3387.000000	NaN	149.000000	NaN	NaN	NaN
75%	5080.000000	NaN	224.000000	NaN	NaN	NaN
max	6766.000000	NaN	300.000000	NaN	NaN	NaN

In [5]: *#Missing Values*

In [5]: `df.isnull().sum()`

```

Out[5]: Request id            0
Pickup point            0
Driver id             2650
Status                0
Request timestamp      0
Drop timestamp        3914
dtype: int64

```

In []: *#Duplicate Values*

In []: `df.duplicated().sum()`

In []: *#7 - Convert Date Columns*

```

In [6]: df['Request timestamp'] = pd.to_datetime(df['Request timestamp'])
df['Drop timestamp'] = pd.to_datetime(df['Drop timestamp'])

```

```
In [ ]: #8 - Create Hour Column
```

```
In [7]: df['Request Hour'] = df['Request timestamp'].dt.hour  
df.head()
```

```
Out[7]:
```

	Request id	Pickup point	Driver id	Status	Request timestamp	Drop timestamp	Request Hour
0	619	Airport	1.0	Trip Completed	2016-11-07 11:51:00	2016-11-07 13:00:00	11
1	867	Airport	1.0	Trip Completed	2016-11-07 17:57:00	2016-11-07 18:47:00	17
2	1807	City	1.0	Trip Completed	2016-12-07 09:17:00	2016-12-07 09:58:00	9
3	2532	Airport	1.0	Trip Completed	2016-12-07 21:08:00	2016-12-07 22:03:00	21
4	3112	City	1.0	Trip Completed	2016-07-13 08:33:00	2016-07-13 09:25:00	8

```
In [ ]: #9 - Status Count
```

```
In [8]: df['Status'].value_counts()
```

```
Out[8]: Status  
Trip Completed      2831  
No Cars Available   2650  
Cancelled           1264  
Name: count, dtype: int64
```

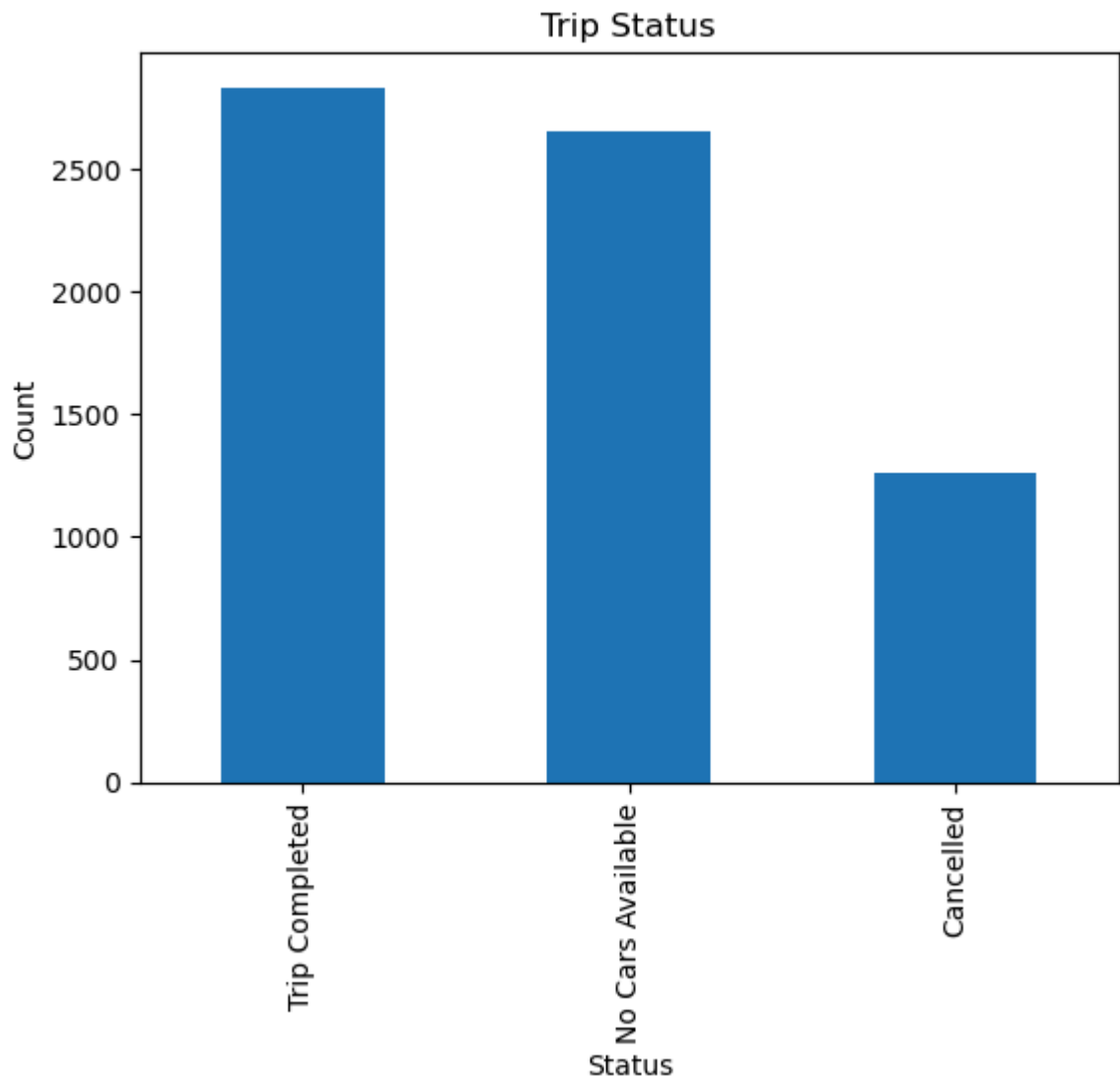
```
In [ ]: #11 - Hour-wise Requests
```

```
In [9]: df['Request Hour'].value_counts().sort_index()
```

```
Out[9]: Request Hour
0      99
1      85
2      99
3      92
4     203
5     445
6     398
7     406
8     423
9     431
10     243
11     171
12     184
13     160
14     136
15     171
16     159
17     418
18     510
19     473
20     492
21     449
22     304
23     194
Name: count, dtype: int64
```

```
In [ ]: 12 - First Visualization
```

```
In [10]: df['Status'].value_counts().plot(kind='bar')
plt.title("Trip Status")
plt.xlabel("Status")
plt.ylabel("Count")
plt.show()
```



Conclusion

From the Exploratory Data Analysis (EDA) of the Uber Request Data, the following conclusions were observed:

- The dataset contains information about Uber requests, pickup points, trip status, and timestamps.
- The majority of requests are either "Trip Completed" or "No Cars Available".
- Demand varies throughout the day, with certain hours experiencing higher request volumes.
- The Airport and City pickup points show different demand and supply patterns.
- The analysis highlights a demand-supply gap during peak hours, leading to cancellations and unfulfilled requests.

Recommendations

- Increase the number of available drivers during peak demand hours.
- Improve driver allocation between the Airport and the City.
- Use historical request data to forecast demand and optimize driver availability.

- Continuously monitor trip status to reduce cancellations and improve customer satisfaction.